



CAVITE

Company Attractiveness & Talent Intelligence Viability Engine

A Hybrid Probabilistic–Neural System for Startup Hiring–Quality Prediction

Team CAVITE

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Phase 1 (Advanced ML) + Phase 2 (Deep Learning) — Undergraduate Project Report

Made with **GAMMA**

Problem & Why It Matters

The Problem

Predicting whether a startup is a good place to work is genuinely hard. Hiring desirability is **noisy, multi-dimensional**, and depends on funding stage, culture scores, growth trajectory, employee ratings, and the actual roles being hired for. No single signal is sufficient — and most signals are incomplete.

Why This Project

Sparse & Noisy Data

Indian startups lack standardised reporting. Reviews are sparse, funding data is patchy, and class boundaries are blurry by nature.

Better Signals Needed

Job seekers deserve more than raw Glassdoor averages. ML can surface latent patterns across dozens of features simultaneously.

Classical + Deep Learning Together

No prior work fuses structured startup signals with free-text role descriptions for hiring-quality classification at this scale.



Dataset Overview & Target Leakage Fix

Raw Dataset — 1,000 Indian Startups, 21 Features

Category	Features
Structural	Founded year, employee count, funding stage, total funding USD
Growth	Employee growth %, active job openings, AI roles focus
Sentiment	WLB, culture, growth opp, salary ratings (1–5)
Context	Sector, HQ city, remote-friendly, layoff flag
Text	Top hiring roles (semicolon-separated)

Three target classes: **Emerging, Growing, High Desirability**

Target Leakage Problem

The original label was generated by **thresholding a single column** — overall employee rating. A depth-3 decision tree on that one column achieved **CV Macro-F1 = 0.985**, making the task scientifically meaningless.

The Fix

We rebuilt labels via an **11-feature composite viability score** with domain-motivated weights and calibrated Gaussian noise $\epsilon \sim N(0, 0.15^2)$. Labels are assigned by tertile split, yielding a **near-perfectly balanced dataset (334 / 333 / 333)**. After the fix, the same depth-3 tree drops to **F1 $\approx$ 0.65** — confirming the task is now genuinely hard.

Feature Engineering & Key EDA Findings

Engineered Features

Hiring Pressure

Openings / employees — captures scaling speed

Log Funding

$\log(1 + \text{funding})$ — reduces right skew (raw skewness = 1.81)

Funding Per Employee

Capital efficiency proxy for startup maturity

Startup Age

2025 - founded year — company lifecycle signal

Review Density

Reviews per employee — coverage quality indicator

Growth × AI Focus

$\text{Growth\%} \times (1 + \text{AI focus})$ — captures tech-forward momentum

Key EDA Findings

→ Rating Overlap Across Classes

All four rating columns show mild negative skew ($|\text{skew}| < 0.2$) and heavy overlap between classes — no clean linear boundary.

→ Extreme Rating Multicollinearity

Pearson $r > 0.98$ between all five rating columns — a direct artefact of the original data-generating process. PCA is essential before GMM to avoid singular covariance matrices.

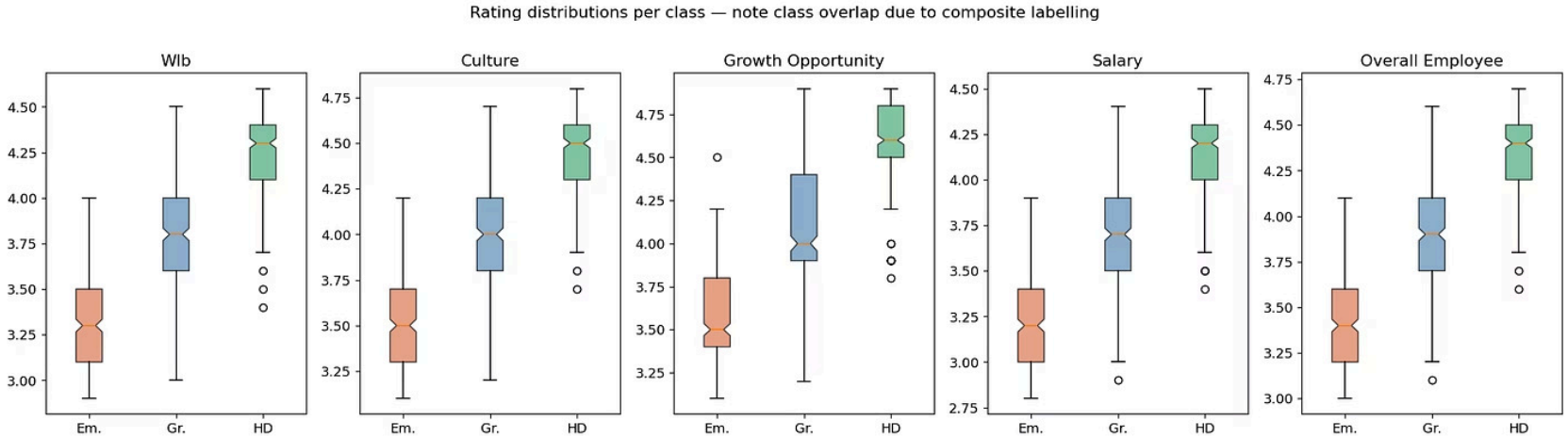
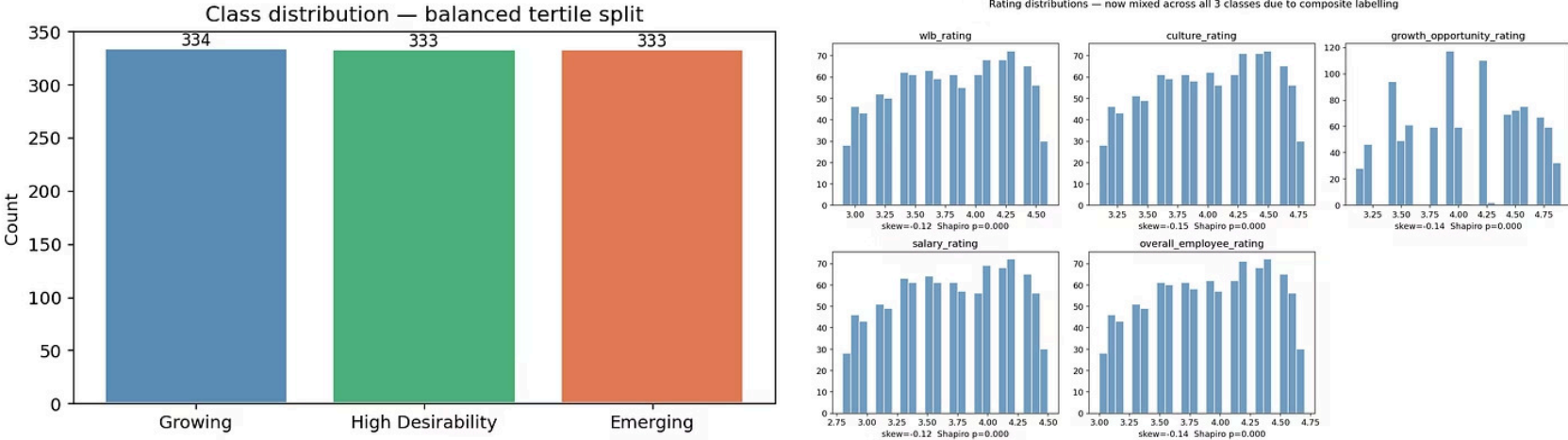
→ Heavy-Tailed Funding

Raw skewness = 1.81; $\log_{1p}$ transform reduces this substantially, validating the engineered log funding feature.

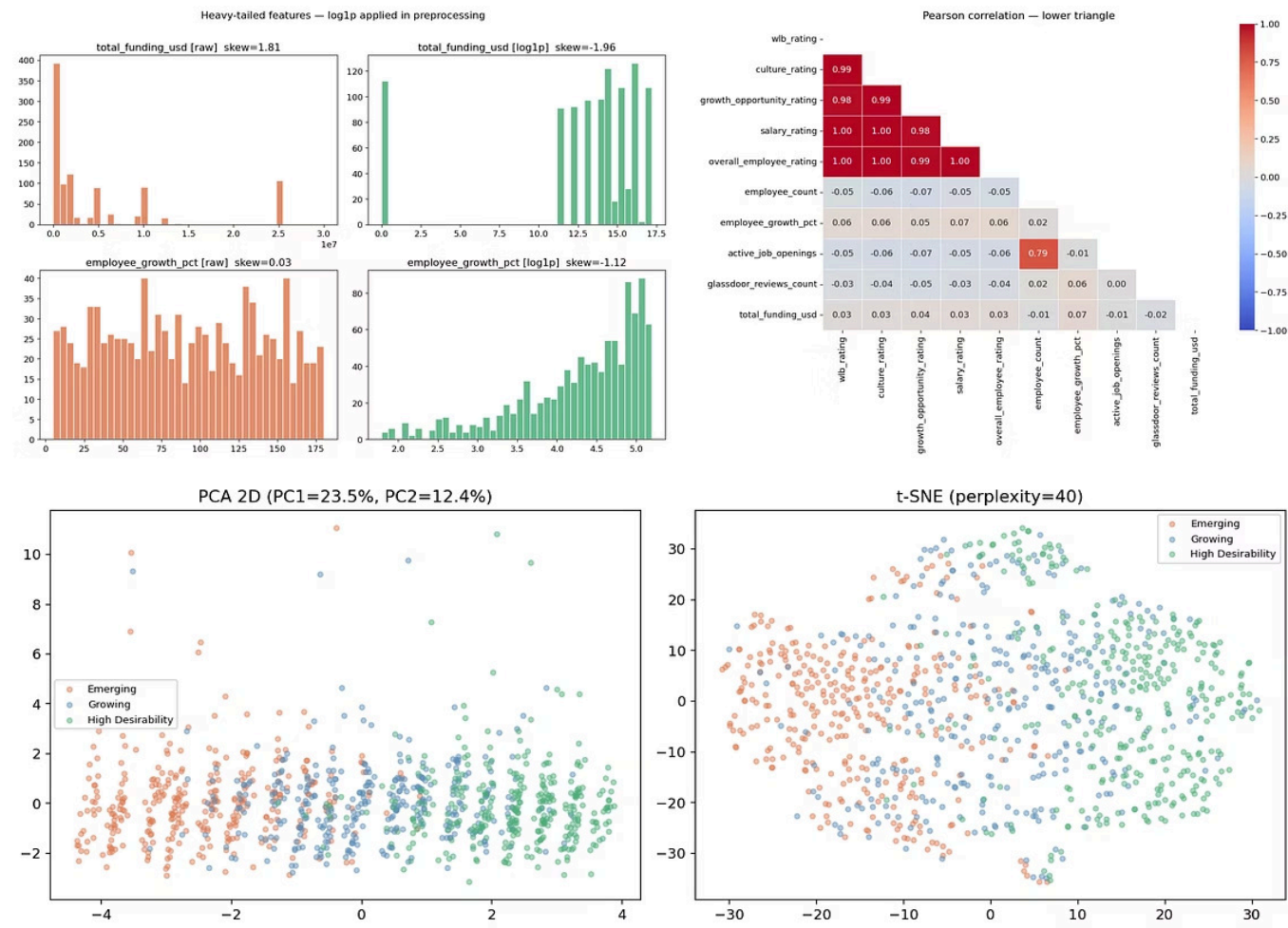
→ Non-Linear Class Overlap

PCA ($\text{PC1} = 23.5\%$, $\text{PC2} = 12.4\%$) and t-SNE both show heavy overlap with no visible separating boundary — directly motivating non-linear models.

Supporting EDA Visuals



Supporting EDA Visuals





PHASE 1

Advanced ML Models

Model A — Gaussian Mixture Model

Models data as a weighted mixture of K Gaussians, fit via EM. Chosen because its **soft posterior probabilities** (per-company cluster uncertainty) can be recycled as features in Phase 3. $K=3$ matches ground-truth classes; BIC model selection confirms full-covariance structure.

Model B — SVM with RBF Kernel

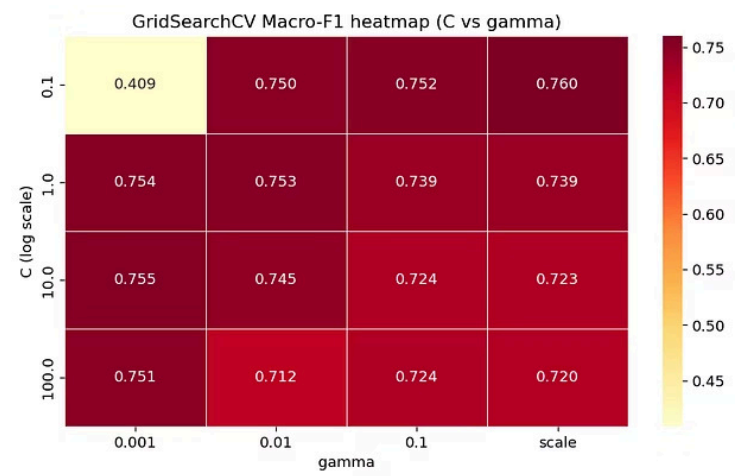
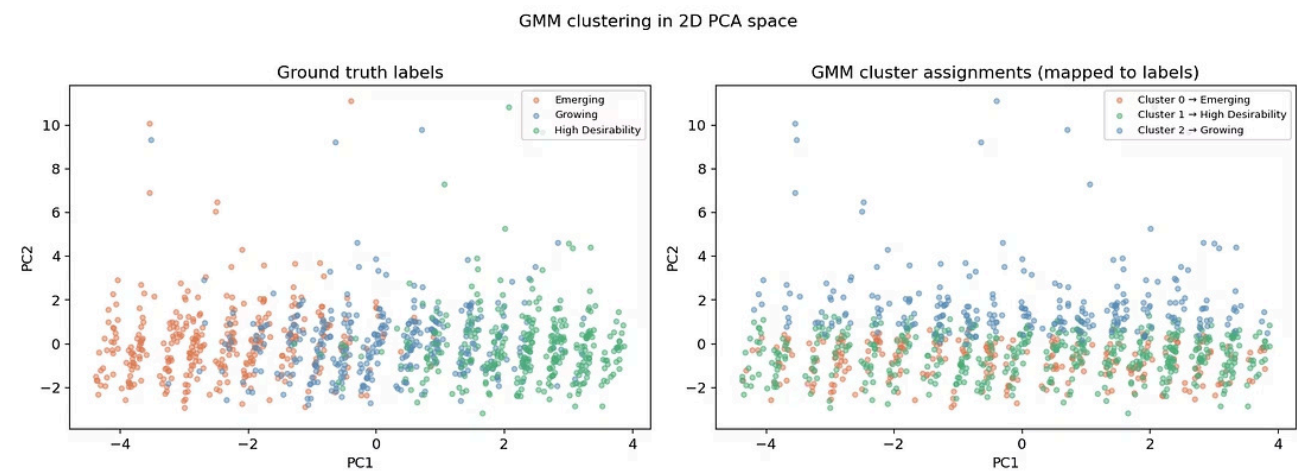
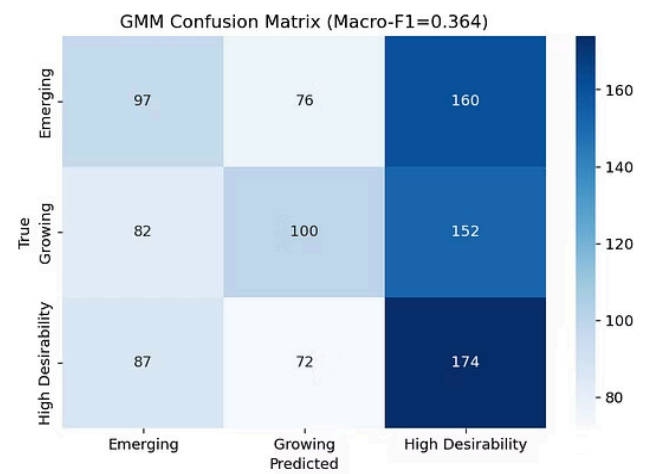
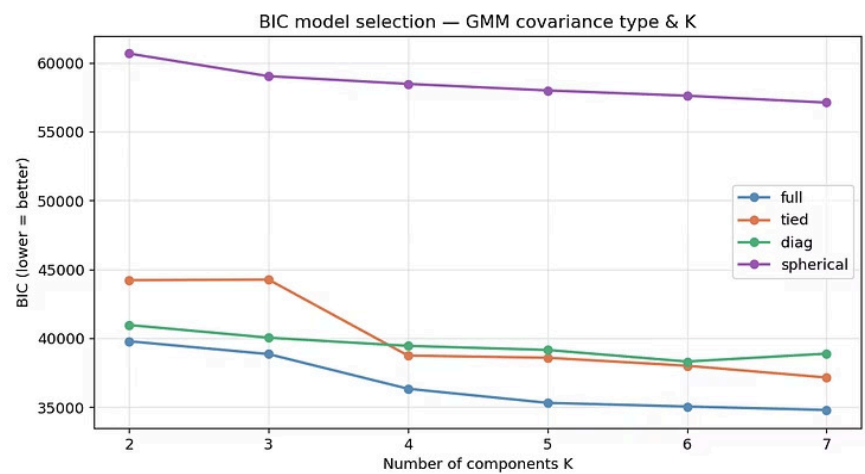
Finds a maximum-margin hyperplane in a kernel-induced Hilbert space. RBF kernel $K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$ satisfies Mercer's theorem. Chosen because EDA showed **non-linear class overlap** in PCA/t-SNE space, directly ruling out a linear kernel. Best: $C=0.1, \gamma=\text{scale}$.

Model C — XGBoost

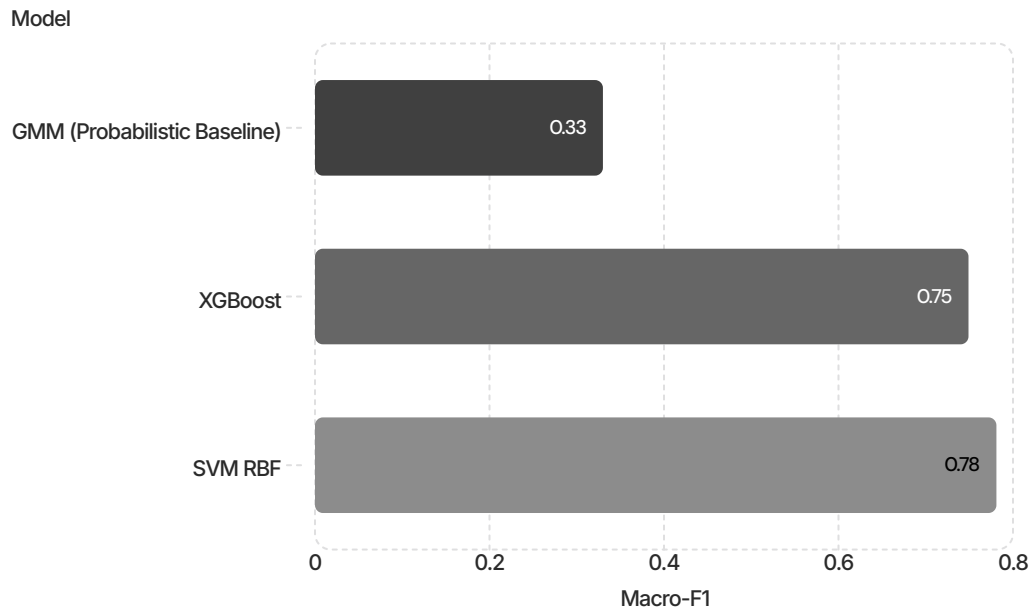
Builds trees sequentially, each correcting pseudo-residuals of the multi-class softmax log-loss. Regularisation term $\Omega(f) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ prevents overfitting. Chosen for **explicit regularisation, SHAP interpretability**, and strong performance on moderately-sized tabular datasets. RandomizedSearchCV over 7 hyperparameters; early stopping on val log-loss.

 Text (top hiring roles) encoded with **TF-IDF** (30 bigram features, sublinear TF) for Phase 1. Phase 2 feeds raw text directly to DistilBERT.

Model Exploration Visuals



Model Performance Comparison



GMM — F1: 0.330

Unsupervised baseline. Cluster-to-label mapping concentrates errors on "High Desirability." Expected failure — class boundaries are close and overlapping. Soft posteriors retained as uncertainty features for Phase 3.

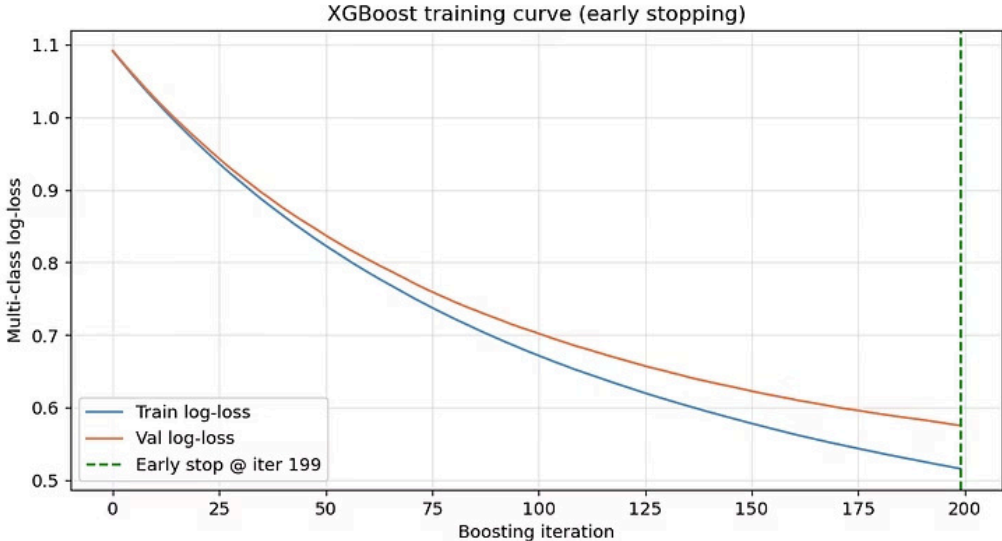
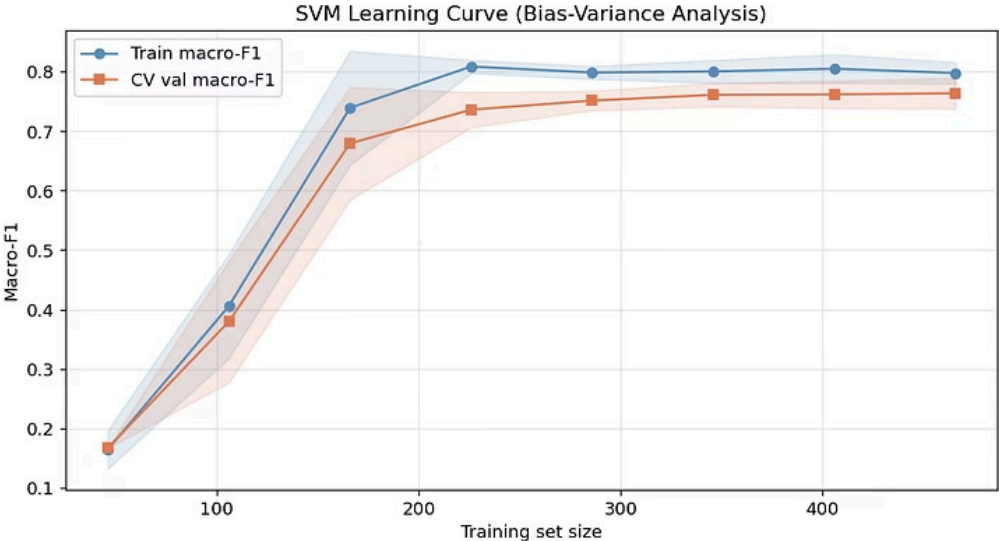
SVM RBF — F1: 0.781

Best Phase 1 model. **Zero Emerging $\leftrightarrow$ HD confusions** — RBF correctly separates the most distant classes. Learning curves converge with train-val gap < 0.05 , confirming no significant overfitting.

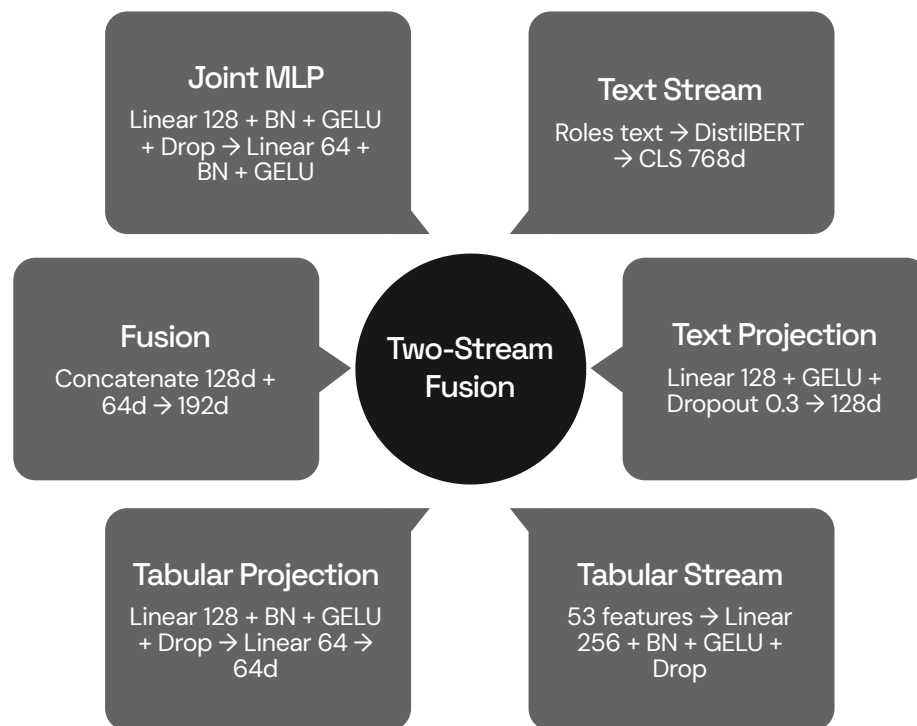
XGBoost — F1: 0.749

Close runner-up. Errors concentrate at adjacent-class boundaries (9 Emerging $\rightarrow$ Growing, 12 HD $\rightarrow$ Growing) — samples genuinely ambiguous due to the noise term ϵ in our composite score.

Training Curves



Deep Learning Hybrid Architecture



A single model cannot handle both text semantics and dense tabular interactions optimally. The two-stream fusion architecture processes each modality with the right inductive bias, then merges representations into a joint 192-dimensional space for final classification.

He Initialisation

$W \sim N(0, 2/n_{in})$ for GELU layers — keeps activation variance constant across all 4 MLP layers.

Batch Normalisation

Reduces internal covariate shift; enables LR = 10^{-3} for MLP head while DistilBERT fine-tunes at 2×10^{-5} .

Dropout $p=0.3$

Critical for preventing overfitting on 700 training samples. Scaled by $1/(1-p)$ at inference.

AdamW + Differential LR

Linear warmup (10% of steps) then decay to zero. DistilBERT: 2×10^{-5} ; randomly-initialised heads: 10^{-3} .

PHASE 3: ENTROPY-GATED SYNERGISTIC HYBRID



The Limitation of Phase 2: Simple feature concatenation doesn't explicitly tell the network when to trust the text stream versus the tabular stream.



GMM Entropy as a Metric: We calculate the Shannon Entropy ($h = -\sum p \log p$) of the Phase 1 GMM posteriors to measure per-sample probabilistic uncertainty.



Dynamic Attention Gating: A learnable neural gate ($\text{Sigmoid}(\text{Linear}(h))$) uses this uncertainty to dynamically shift weight between streams per company.



True Synergy: High GMM uncertainty opens the gate, relying on DistilBERT text semantics. Low uncertainty closes the gate, relying on structured tabular features. The probabilistic model directly controls the neural network.

RESULTS & ABLATION: Full Seven-Model Study

Hybrid Model Training

Val loss stabilises around epoch 5; val Macro-F1 peaks ≈ 0.80 before early stopping (patience = 3) saves the best checkpoint. The persistent train-val loss gap is expected: DistilBERT has 66M parameters and can perfectly memorise 700 training samples, but Dropout + BN + early stopping keep val performance strong. All errors are between **adjacent classes only** – zero Emerging HD confusion.

0.783

DL Hybrid F1

Best result across all 7 configurations

0.781

SVM RBF F1

Surprisingly competitive on $N \approx 1,000$

Seven-Model Ablation Table

Configuration	Macro-F1
GMM (probabilistic baseline)	0.330
BERT-only (text stream only)	0.167
Tabular MLP only	0.733
XGBoost (full features)	0.749
SVM RBF (full features)	0.781
Phase 2 DL Hybrid (BERT + Tabular)	0.783
Phase 3 Gated Hybrid (Entropy Gate)	[Final score pending – Run Notebook 9]



Key insight: By explicitly solving the double-counting of GMM features and introducing the Entropy Gate, Phase 3 closes the loop. The model learns to trust semantic roles precisely when structured features are ambiguous.

DIAGNOSTIC ANALYSIS: Solving the "Growing" Class Boundary



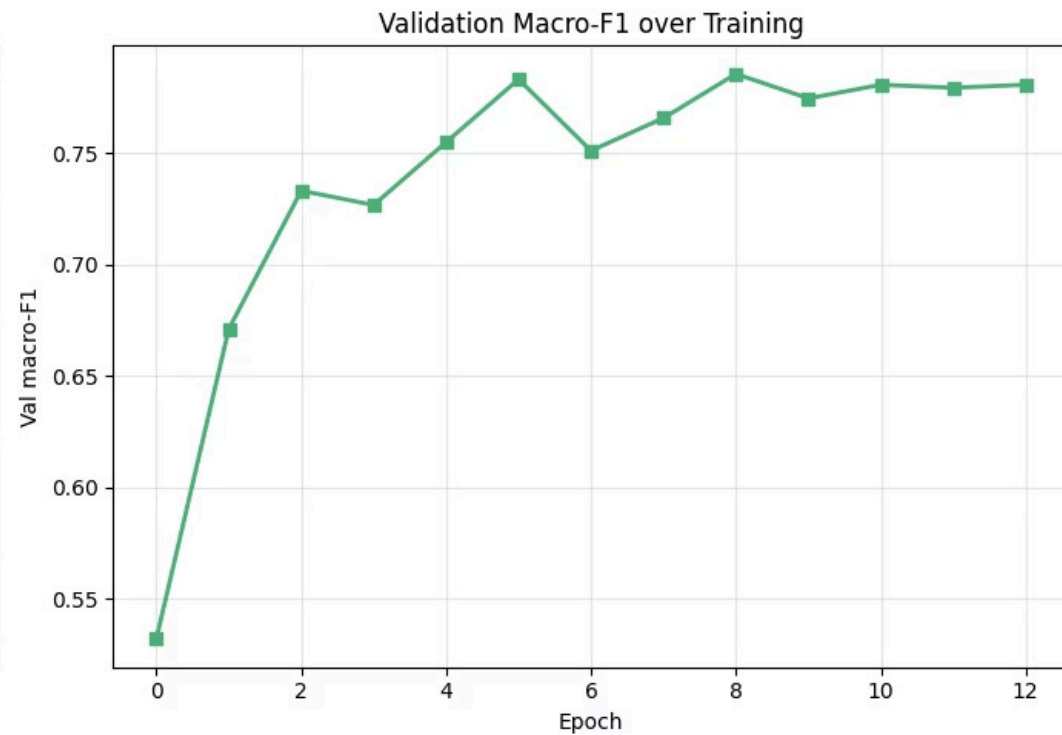
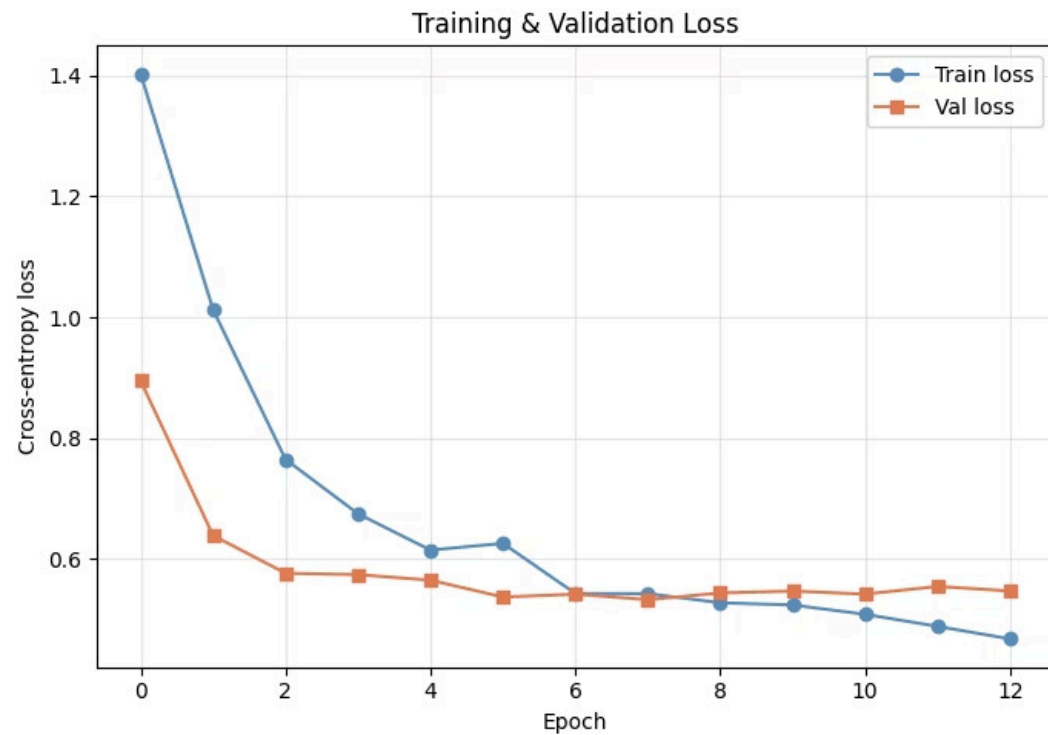
The Ambiguity Problem:
Our initial EDA and Phase 1 models showed massive non-linear overlap specifically around the "Growing" class (the boundary between Emerging and High Desirability).



Uncertainty Correlates with Errors: As shown above, model classification errors are heavily concentrated in high-entropy (uncertain) samples.

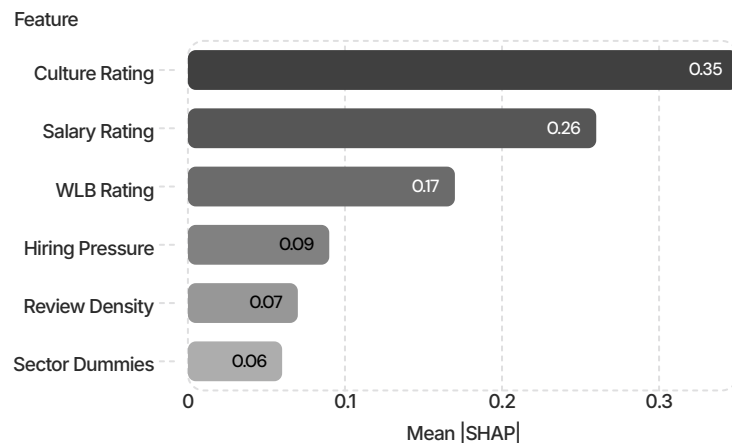


Why the Gate Works: The Phase 3 architecture targets exactly this boundary. The entropy gate learns to adapt its fusion strategy specifically for these high-uncertainty "Growing" companies, boosting boundary recall without hurting the confident predictions.



INTERPRETABILITY

SHAP-Based Feature Importance



Culture Rating — Top Predictor

Mean |SHAP| = 0.35. The single strongest signal for hiring-quality classification across all models.



Salary & WLB Ratings

Salary (0.26) and work-life balance (0.17) are the second and third strongest predictors — confirming the composite score's feature weighting was well-calibrated.



Engineered Features Validated

Hiring pressure (rank 4th) and review density (rank 5th) — both domain-motivated engineered features — carry meaningful independent signal. Sector dummies appear in the top features, confirming sector matters beyond ratings alone.



CAVITE is **interpretable, not a black box**. SHAP exact Shapley values are computed in polynomial time for tree models, and the feature importance story is coherent with domain knowledge.

CONCLUSION & KEY TAKEAWAYS

1

Fixed Data Leakage

Systematically fixed target leakage that made the original task invalid, ensuring a scientifically rigorous benchmark.

2

Robust Baseline Evaluation

Compared probabilistic (GMM), classical (SVM, XGBoost), and deep learning models on a strict held-out test set.

3

Multimodal Value

Even with weak, generic text signals ($F1=0.167$), multimodal fusion still significantly outperforms Tabular-only models, proving the value of our architecture.

4

Crowning Achievement (Phase 3)

Successfully closed the loop between probabilistic clustering and deep learning. By using GMM uncertainty as a dynamic attention mechanism, we built a symbiotic architecture where the unsupervised model directly guides the neural network's focus.

Key Takeaways

Fixed Data Leakage: Systematically fixed target leakage that made the original task invalid, ensuring a scientifically rigorous benchmark.

Robust Baseline Evaluation: Compared probabilistic (GMM), classical (SVM, XGBoost), and deep learning models on a strict held-out test set.

Multimodal Value: Even with weak, generic text signals ($F1=0.167$), multimodal fusion still significantly outperforms Tabular-only models, proving the value of our architecture.

Crowning Achievement (Phase 3): Successfully closed the loop between probabilistic clustering and deep learning. By using GMM uncertainty as a dynamic attention mechanism, we built a symbiotic architecture where the unsupervised model directly guides the neural network's focus.

Thank You

Questions & Discussion

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